

The impact of generalized anxiety disorder and stressful life events on risk for major depressive episodes

JOHN M. HETTEMA^{1*}, JONATHAN W. KUHN¹, CAROL A. PRESCOTT¹
AND KENNETH S. KENDLER^{1,2}

¹ *Virginia Institute for Psychiatry and Behavioral Genetics and Departments of Psychiatry and*

² *Human Genetics, Virginia of Virginia Commonwealth University, Richmond, VA, USA*

ABSTRACT

Background. Both generalized anxiety disorder (GAD) and stressful life events (SLEs) are established risk factors for major depressive disorder, but no studies exist that examine the inter-relationship of their impact on depressive onsets. In this study, we sought to analyze the joint effects of prior history of GAD and recent SLEs on risk for major depressive episodes, comparing these in men and women.

Method. In a population-based sample of 8068 adult twins, Cox proportional hazard models were used to predict onsets of major depression from reported prior GAD and last-year SLEs rated on long-term contextual threat.

Results. For all levels of threat, prior GAD increases risk for depression, with a monotonic relationship between threat level and risk. While females without prior GAD consistently show higher depressive risk than males, this is no longer the case in subjects with prior GAD who have experienced SLEs. Rather, **males appear to be more vulnerable to the depressogenic effects of both prior GAD and SLEs.**

Conclusion. The effects of prior GAD and SLEs jointly increase the risk of depression in both sexes, but disproportionately so in males.

INTRODUCTION

Major depressive disorder (MDD) is highly co-morbid with all anxiety disorders (Maser & Cloninger, 1990) particularly generalized anxiety disorder (GAD; Judd *et al.* 1998). The onset of most anxiety disorders precede that of MDD, suggesting their role as risk factors for MDD (Wittchen *et al.* 2000). In fact, the risk of depression in individuals with chronic anxiety disorders like GAD is greater than the risk of anxiety in individuals with depressive disorders (Angst *et al.* 1990). Two large population-based

studies found that the prior existence of any anxiety disorder significantly increases the risk for subsequent development of MDD, with the strongest risk attributable to GAD (Kessler *et al.* 1996; Hettema *et al.* 2003). Furthermore, using a longitudinal epidemiologic sample, Breslau *et al.* (1995) found that a pre-existing anxiety disorder significantly increased the risk of MDD and may account in part for the greater prevalence of MDD in females. Similar to the effects of anxiety disorders, stressful life events (SLEs) significantly contribute to the risk of onset of MDD (Kessler, 1997; Kendler *et al.* 1999). The aim of this study was to examine the joint effects of prior history of GAD and recent SLEs on risk for onset of an episode of MDD, exploring the potential for gender to modify these effects.

* Address for correspondence: John M. Hettema, M.D., Ph.D.,
Department of Psychiatry, PO Box 980126, Richmond, VA
23298-0126.
(Email: jhettema@hsc.vcu.edu)

METHOD

Sample

The twins in this study derive from the population-based Virginia Twin Registry (Kendler & Prescott, 1999) which now constitutes part of the Mid-Atlantic Twin Registry. Female–female (FF) twin pairs, from birth years 1934–1974, became eligible if both members previously responded to a mailed questionnaire in 1987–1988, the response rate to which was 64%. They have been approached for up to four subsequent waves of personal interviews (either telephone or face-to-face) from 1988 to 1997, with cooperation rates ranging from 85% to 92%. The male–male and male–female (MMMF) twin pairs, covering the birth years 1940–1974 were ascertained in a separate study – with an initial cooperation rate of 72.4% – and have been approached for two waves of personal interviews from 1993 until 1998. The mean interval (months) between the interview waves are as follows: FF1–FF2 (17.0), FF2–FF3 (44.3), FF3–FF4 (31.1), MMMF1–MMMF2 (19.0). After an explanation of the research protocol, signed informed consent was obtained prior to all face-to-face interviews and verbal informed consent prior to all telephone interviews.

Assessments

During each interview wave, we assessed the occurrence over the last year of 14 individual symptoms which represented the disaggregated nine ‘A Criteria’ for MDD in DSM-III-R (APA, 1987). For each reported symptom, interviewers probed to insure that it was due neither to physical illness nor medication. The respondents and interviewers then aggregated symptoms reported over the last year into co-occurring syndromes. If depressive syndromes occurred, respondents were asked when they occurred and the months of their onset and offset. **To be eligible for the occurrence of a new depressive episode, the respondent had to report being basically symptom-free and being ‘back to his/her normal self’ for at least 2 weeks.** The diagnosis of MDD was made by computer algorithm incorporating the DSM-III-R criteria, except criterion B2 (which excludes ‘uncomplicated bereavement’).

Lifetime diagnosis of GAD followed DSM-III-R criteria including diagnostic hierarchy

with MDD but employed a minimum duration of 1 month instead of 6 months to increase prevalence and, hence, statistical power, as prior analyses in this sample suggest that GAD with these two duration criteria lie on the same dimension of liability (Hettema *et al.* 2001). **Subjects were assigned a diagnosis of ‘prior GAD’ if they met criteria for GAD as above and reported their date of onset as occurring at least 1 year before the date of interview,** such that onset occurred prior to the period for which we assessed SLEs and MDD onsets. Similarly, we also defined ‘current GAD’ for periods during the last year when a subject met criteria for GAD *in the month that a SLE occurred*, differentiating this potential ‘state effect’ from ‘trait effects’ related to prior GAD.

Our interviews assessed, to the nearest month during the last 1-year period, the occurrence of **11 ‘personal’ stressful life events (events occurring primarily to the informant): ‘assault’, ‘robbed’, ‘divorce/separation’, ‘serious marital problems’, ‘major financial problem’, ‘serious housing problems’, ‘serious illness or injury’, ‘job loss’, ‘serious difficulties at work’, ‘legal problems’, and ‘loss of confidant’.** We also assessed four classes of ‘network’ events, meaning events that primarily involved an individual in the respondent’s social network. Network was here defined as spouse, child, parent, sibling, other close relative or ‘someone else close to you’. These event classes consisted of the following: ‘getting along with’ (serious trouble getting along with an individual in the network), ‘crisis’, ‘death’, or ‘illness’ of someone in the network.

Each SLE reported in waves 3 and 4 of the FF sample and wave 2 of the MMMF sample was rated by the interviewer on the level of long-term contextual threat (LTCT), where ‘long-term’ means persisting at least 10–14 days after the event. Following Brown, we instructed our interviewers to rate ‘what most people would be expected to feel about an event in a particular set of circumstances and biography, taking no account either of what the respondent says about his or her reaction or about any psychiatric or physical symptoms that followed it’ (Brown, 1989, p. 24). LTCT was rated on a 4-point scale: minor, low moderate, high moderate, and severe. Reliability of our ratings of LTCT was determined by inter-rater and

test–retest designs. Inter-rater reliability was assessed by having experienced interviewers review tape recordings of the interview sections in which 92 randomly selected individual SLEs were evaluated. Test–retest reliability was obtained by repeating the interview with 191 respondents at a mean interval of 4 weeks. We obtained 173 scored life events that were reported to have occurred within 1 month of each other and, we assumed, represented the same event. We assessed reliability by Spearman correlation (r_s) and weighted kappa (κ) (Fleiss *et al.* 1969). The test–retest reliability for LTCT was $r_s = +0.60$ and $\kappa = +0.41$, while inter-rater reliability was $r_s = +0.69$ and $\kappa = +0.67$.

Analysis

Using person-months as the unit of analysis, these analyses were conducted with a Cox proportional hazards model operationalized in the SAS procedure PHREG (Cox, 1972). Three categories of predictor variables were used: prior GAD, sex and LTCT. When multiple events occurred in the same month, LTCT was coded as the highest threat level of any recorded event. The dependent variable was the onset of a depressive episode. In the models, the data were stratified in an effort to control for within-subject correlation across interview waves and depressive onsets. Each strata consisted of data for subjects for one study each with a specific number of onsets – specifically zero, one or two onsets in the past 13 months for subjects from three studies, totaling nine strata (three waves with up to onset designations each). There were too few subjects with three or more onsets to include such data.

LTCT was coded so that 0 meant no SLE occurrence in the month and 1–4 meant the occurrence of a SLE with minor, low moderate, high moderate and severe LTCT. We used thermometer coding (Masters, 1993) to incorporate this ordinal structure and simplify interpretation of interactions. For example, if effects are seen at the level of minor LTCT (= 1), this signifies a difference between no SLE (= 0) and any level of LTCT (1–4). If effects are seen at LTCT levels 1, 2, and 3, the hazard ratios (HRs) are multiplicatively cumulative with this modeling structure. The SLEs were then included as time-dependent covariates with a linear decay over 3 months. The final model was

Table 1. *Results of univariate Cox modeling for the individual effects of sex, prior generalized anxiety disorder (prior GAD), and stressful life events scored by long-term contextual threat (LTCT) to predict hazard ratio for onset of major depressive episodes*

Model parameter	Hazard ratio	95% CI	χ^2 (df = 1)	p value
Sex	1.62	1.36–1.92	30.9	<0.0001
Prior GAD	2.14	1.84–2.50	95.5	<0.0001
Mild LTCT	1.40	1.14–1.72	10.4	0.0013
Low moderate LTCT	1.68	1.33–2.10	20.1	<0.0001
High moderate LTCT	2.06	1.76–2.43	76.9	<0.0001
Severe LTCT	2.21	1.87–2.62	84.8	<0.0001

CI, Confidence interval.

found by starting with all main effects and two- and three-way interactions for the predictor variables and removing terms non-significant at the $p < 0.05$ level by step-wise elimination.

Information for these analyses comes from a total of 8068 individuals (4240 males and 3828 females) who participated in waves 3 and 4 of the FF study and wave 2 of the MMMF study. These individuals reported a total of 1089 onsets of MDD episodes for 10 501 periods of observation where each period of observation either began at the start of a 1-year prevalence window or at the time of recovery from an episode and ended either at the time of onset of a depressive episode or at the conclusion of that 1-year window. **The rates of broadly defined prior GAD were 15.4% in females and 7.3% in males. The 12-month rate for MDD onsets (limited to one onset per individual) is 8.9% in females and 7.0% in males, and 7.2% in those without prior GAD and 17.4% in those with prior GAD.**

RESULTS

Univariate Cox modeling results for the effects of the main predictor variables on risk for MDD onsets are displayed in Table 1. The individual effects from all of these are highly significant for predicting onsets of MDD in our sample. **Female sex, prior GAD, and each level of SLE adversity coded as LTCT independently carry increased risk, quantified as a HR, in the range 1.4–2.2.**

Table 2 lists the results of the best-fit multivariate Cox model predicting onsets of MDD.

Table 2. *Best-fit multivariate Cox model including all significant main effects and 2- and 3-way interactions for sex, prior generalized anxiety disorder (prior GAD), and stressful life events scored by long-term contextual threat (LTCT) to predict hazard ratio for onset of major depressive episodes*

Model parameter	Hazard ratio	95% CI	χ^2 (df=1)	p value
Sex	2.89	2.23–3.76	63.1	<0.0001
Prior GAD	2.57	1.99–3.29	54.5	<0.0001
Mild LTCT	2.19	1.71–2.82	38.3	<0.0001
Low moderate LTCT	1.67	1.33–2.09	19.8	<0.0001
High moderate LTCT	2.09	1.78–2.46	79.2	<0.0001
Severe LTCT	2.27	1.91–2.68	89.9	<0.0001
Sex-by-prior GAD	0.59	0.43–0.82	10.4	0.0013
Sex-by-(mild or higher LTCT)	0.48	0.38–0.59	43.5	<0.0001

CI, Confidence interval.

As in the univariate results, there are significant main effects of each of the predictor variables. The HR is coded with baseline risk of 1 assigned to a male without prior GAD and LTCT=0 (no SLE), with all other groups referenced to this. When controlling for the effects of the other predictors, women carry a baseline risk of 2.89 compared to men, increased from 1.62 in the univariate results. The effect sizes of the other predictors are similar to those found in Table 1. **A prior history of GAD increases a subject's risk for depressive onset by a HR of 2.57 compared to no prior GAD.** Each increasing level of LTCT carries an additional risk, with HRs ranging from 1.7–2.3. **Since these are multiplicative, the HR for a depressive onset associated with a SLE rated as LTCT of 1, 2, 3, or 4 equals 2.19, 3.66, 7.64, and 15.97 respectively.**

In addition, there are significant interactions between sex and prior GAD as well as between sex and LTCT levels of mild severity or higher. **These are negative interactions with HRs <1, signifying a downward adjustment of the joint effects of both variables compared to that predicted by their main effects alone.** Thus, even though prior GAD and SLEs robustly increase risk of depressive onset in both sexes, **women are less sensitive to the depressogenic effects of both SLEs (LTCT ≥ 1) and prior GAD than are men.** For example, once one accounts for the baseline risk associated with being female

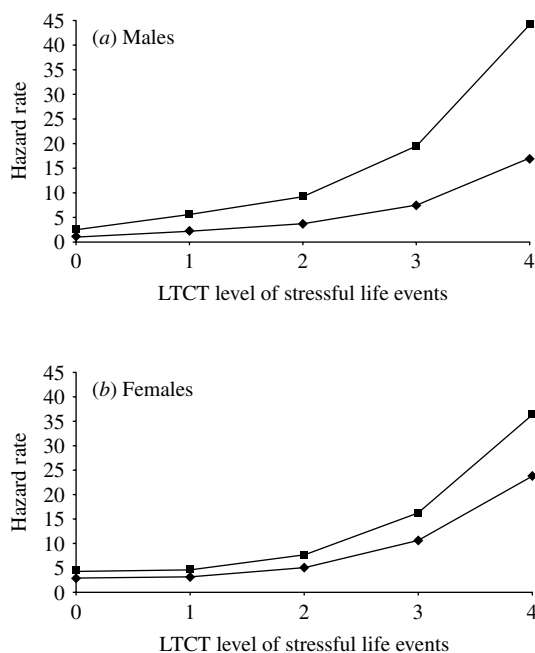


FIG. 1. Hazard ratios indicating risk for onset of major depression as a function of long-term contextual threat level (LTCT) of stressful life events for (a) males and (b) females (total $n=8068$). Curves for subjects without prior history of generalized anxiety disorder (prior GAD) indicated by diamonds (◆) and those with prior history of GAD indicated by squares (■). The hazard rate is standardized to unity for a male with no prior GAD and no exposure to stressful life events.

(HR=2.89), **the effect of prior GAD is 0.59 times smaller in women than in men**, although the combined risk of being female and having prior GAD is $(2.89) \times (0.59) = 1.70$ times that of a male with prior GAD at LTCT=0 (no SLE). Similarly, the effect of SLEs with LTCT ≥ 1 is 0.48 times smaller in women, **though women with any level SLEs are at $(2.89) \times (0.48) = 1.39$ higher risk for a depressive onset than men.** There are no significant three-way interactions.

Figs 1 and 2 graph the predictions of the best-fit model as a function of LTCT level. Fig. 1 compares the hazard rates for individuals with and without prior GAD respectively, separately in men and women. Two key findings derive from these graphs. First, **risk increases steeply with adversity level for both genders.** Second, **there is a substantial increase in risk associated with prior history of GAD at all levels of LTCT, and this increase is more pronounced in males than females.**

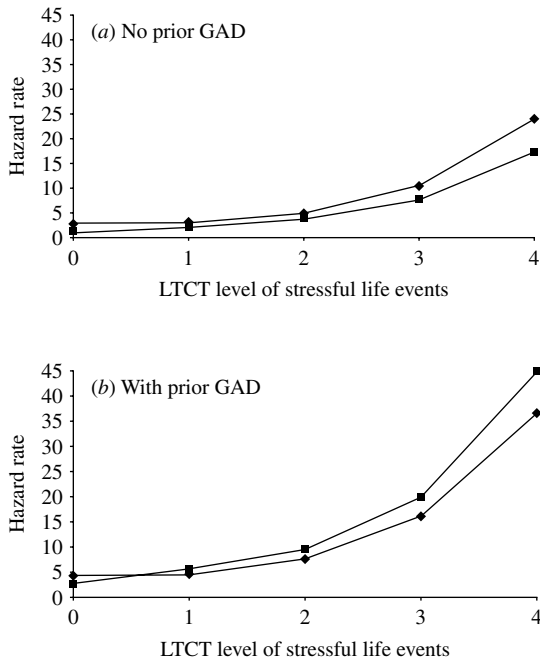


FIG. 2. Hazard ratios indicating risk for onset of major depression as a function of long-term contextual threat level (LTCT) of stressful life events for those without (a) and those with (b) prior history of generalized anxiety disorder (prior GAD) (total $n=8068$). Curves for female subjects indicated by diamonds (♦) and male subjects indicated by squares (■). The hazard rate is standardized to unity for a male with no prior GAD and no exposure to stressful life events.

Fig. 2 directly compares the hazard rates in males and females separately for those without and with prior GAD respectively. In subjects without prior GAD, risk for depressive onset is greater in women than men at all levels of adversity. For those with prior GAD, while women's risk exceeds that of men who have not experienced a SLE prior to their depressive onset (LTCT=0), this is not the case once the effects of SLEs are taken into account. Rather, the excess risk in females seen with no SLEs disappears as adversity level rises. (Although it may appear that male risk exceeds that of females at high LTCT, the values are not significantly different.) In fact, a model that includes only sex, prior GAD, and their interaction (not shown) predicts that men and women with prior GAD are at overall (i.e. 'averaged' over LTCT level) equal risk for onset of a depressive episode. Thus, prior GAD 'equalizes'

the depressive risk between men and women in our models.

DISCUSSION

In the present study, we have examined the joint effects of SLEs and prior history GAD on the risk for major depressive onset. In our best-fit model, both of these factors significantly and independently predicted risk in men and women. In addition, there were significant interactions between each of these factors and sex, producing sex differences in their effects. Three main findings derive from these results. First, for both for subjects with and without prior GAD, increasing adversity level predicts higher risk for MDD. Second, for every level of threat, prior GAD predicts increased risk for MDD. Thus, prior GAD is associated with greater risk of depressive onset due to the impact of SLEs, suggesting that GAD sensitizes an individual to the depressogenic effects of stress. Finally, the female-to-male HR for MDD decreases as adversity level increases. In particular, while females without prior GAD have consistently higher risk for MDD than males at every level of threat, this is not the case for subjects with prior GAD. Rather, males with GAD have lower risk for MDD than females only at low levels of adversity, with their risk becoming comparable to that of females as the level of adversity rises. This appears to be due to the higher sensitivity of men to depressive onsets due to both prior GAD and SLEs. Thus, males and females with GAD are at similarly high risks of developing MDD after experiencing SLEs.

One might speculate on how prior GAD itself is associated with increased risk for depression as well as sensitizes an individual to the depressogenic effects of SLEs. These may not be distinct questions. Previous studies in our twin sample have estimated that GAD and MDD share a significant portion of their genetic risk (Kendler *et al.* 1992). Therefore, having an elevated genetic loading for GAD and MDD puts those with GAD at higher risk for depressive onsets at baseline as well as in response to SLEs (Kendler *et al.* 1995). Another large population-based twin study that examined the relationship between the personality trait neuroticism and self-report symptoms of anxiety

and depression concluded that genetic variation in these symptoms is largely dependent on the same factors as those affecting neuroticism (Jardine *et al.* 1984). In addition, our group has demonstrated that neuroticism possesses high correlation in genetic risk with both GAD (Hettema *et al.* 2004) and MDD (Fanous *et al.* 2002). Therefore, one model for the development of GAD and MDD is the following: neurotic traits index one's genetic susceptibility to either GAD or MDD, with other factors (modifier genes or environmental factors such as SLEs) determining the often observed sequence of GAD predating MDD. In fact, a recent study in this sample examined the relationship between neuroticism and SLEs on onsets of depression, finding a significant but somewhat lower HR for neuroticism predicting MDD than we found for GAD in the current study (Kendler *et al.* 2004). Thus, individuals who have already developed GAD most likely have more risk factors on average than those with higher neuroticism scores without GAD and are further along on such a pathophysiological progression, making GAD a more proximal and potent risk factor for MDD. The clinical relevance of this is supported by one study that found that pharmacologic treatment of GAD decreases the risk for MDD (Goodwin & Gorman, 2002).

We are aware of no prior studies specifically examining the joint effects of GAD and life events on risk for onsets of depression. Several prior studies have investigated how gender modifies the depressogenic effects of SLEs. While we found evidence for a negative interaction between sex and LTCT of ≥ 1 indicating an overall decreased sensitivity of women to SLEs in this and a prior analysis, most other studies have found either no overall gender differences or increased sensitivity of women either globally or to specific classes of SLEs (Bebbington, 1998). The reasons for such discrepant results are unclear. We also performed *post-hoc* analyses, comparing the effect of 'personal' life events to that of 'network' life events on depressive risk. There were no differences in the overall pattern of results.

Few studies have found evidence for gender-specific effects of prior anxiety on risk for depression. Breslau *et al.* (1995) reported that pre-existing anxiety disorders significantly

increase the risk of lifetime MDD and may account for part of the observed sex difference in prevalence. In this study and others from our sample (Hettema *et al.* 2003; Kendler *et al.* 2004), we could not reproduce that finding. However, Breslau *et al.* did report that the interaction term between sex and prior anxiety, though not significant ($p > 0.30$), favored the interpretation that males with prior anxiety are more vulnerable than females to depression, consistent with our findings.

Limitations

These results should be considered in the context of several potentially significant methodologic limitations. First, our analyses assumed that SLEs preceded depressive onsets when reported in the same month. While our data is retrospective (over the recall interval of up to 1 year), and we have no external validation of the reports of SLEs or depressive onsets, other analyses in this sample (Kendler *et al.* 2004) support the assumption that when both onsets and SLEs are reported in the same month, the SLE precedes the onset the vast majority of the time.

By including history of GAD that existed prior to the 1-year window of life events and depressive onsets that we studied, we were able to demonstrate that there are persistent residual or 'trait effects' on risk for MDD. We also attempted to model the 'state effects' of GAD concurrent with the timing of the SLE, but such life events that occurred during a GAD episode were too few, despite the size of our sample, to provide sufficient power to detect such effects. In fact, when we deleted those 28 events from the current study, the model did not change substantially.

We used modified diagnostic criteria to define prior GAD in order to increase prevalence and, hence, statistical power, in our models. When we re-analyzed the data using 6-month instead of 1-month duration criteria, the overall results were unchanged except that the hazard rate for the interaction between sex and prior GAD was of similar magnitude but no longer statistically significant.

Finally, we performed these analyses in a population-based sample of Caucasian twins in Virginia, and therefore, these results may not generalize to other populations.

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DECLARATION OF INTEREST

None.

REFERENCES

- APA (1987). *Diagnostic and Statistical Manual of Mental Disorders* (revised 3rd edn). American Psychiatric Association: Washington, DC.
- Angst, J., Vollrath, M., Merikangas, K. & Ernst, C. (1990). Comorbidity of anxiety and depression in the Zurich Cohort Study of Young Adults. In *Comorbidity of Mood and Anxiety Disorders* (ed. J. D. Maser and C. R. Cloninger), pp. 123–137. American Psychiatric Press, Inc: Washington, DC.
- Bebbington, P. E. (1998). Editorial: Sex and depression. *Psychological Medicine* **28**, 1–8.
- Breslau, N., Schultz, L. & Peterson, E. (1995). Sex differences in depression: a role for preexisting anxiety. *Psychiatry Research* **58**, 1–12.
- Brown, G. W. (1989). Life events and measurement. In *Life Events and Illness* (ed. G. W. Brown and T. O. Harris), pp. 3–45. Guilford Press: New York.
- Cox, D. R. (1972). Regression models and life tables (with discussion). *Journal of the Royal Statistical Society (Series B)* **34**, 187–220.
- Fanous, A., Gardner, C. O., Prescott, C. A., Cancro, R. & Kendler, K. S. (2002). Neuroticism, major depression and gender: a population-based twin study. *Psychological Medicine* **32**, 719–728.
- Fleiss, J. S., Cohen, J. & Everett, B. S. (1969). Large sample standard errors of kappa and weighted kappa. *Psychological Bulletin* **72**, 323–327.
- Goodwin, R. D. & Gorman, J. M. (2002). Psychopharmacologic treatment of generalized anxiety disorder and the risk of major depression. *American Journal of Psychiatry* **159**, 1935–1937.
- Hettema, J. M., Prescott, C. A. & Kendler, K. S. (2001). A population-based twin study of generalized anxiety disorder in men and women. *Journal of Nervous and Mental Disease* **189**, 413–420.
- Hettema, J. M., Prescott, C. A. & Kendler, K. S. (2003). The effects of anxiety, substance use and conduct disorders on risk of major depressive disorder. *Psychological Medicine* **33**, 1423–1432.
- Hettema, J. M., Prescott, C. A. & Kendler, K. S. (2004). Genetic and environmental sources of covariation between generalized anxiety disorder and neuroticism. *American Journal of Psychiatry* **161**, 1581–1587.
- Jardine, R., Martin, N. G. & Henderson, A. S. (1984). Genetic covariation between neuroticism and the symptoms of anxiety and depression. *Genetic Epidemiology* **1**, 89–107.
- Judd, L. L., Kessler, R. C., Zeller, P. V., Wittchen, H.-U. & Kunovac, J. L. (1998). Comorbidity as a fundamental feature of generalized anxiety disorders: results from the National Comorbidity Study (NCS). *Acta Psychiatrica Scandinavica* **98**, 6–11.
- Kendler, K. S., Karkowski, L. M. & Prescott, C. A. (1999). Causal relationship between stressful life events and the onset of major depression. *American Journal of Psychiatry* **156**, 837–841.
- Kendler, K. S., Kessler, R. C., Walters, E. E., MacLean, C. J., Sham, P. C., Neale, M. C., Heath, A. C. & Eaves, L. J. (1995). Stressful life events, genetic liability and onset of an episode of major depression in women. *American Journal of Psychiatry* **152**, 833–842.
- Kendler, K. S., Kuhn, J. & Prescott, C. A. (2004). The interrelationship of neuroticism, sex, and stressful life events in the prediction of episodes of major depression. *American Journal of Psychiatry* **161**, 631–636.
- Kendler, K. S., Neale, M. C., Kessler, R. C., Heath, A. C. & Eaves, L. J. (1992). Major depression and generalized anxiety disorder: same genes, (partly) different environments? *Archives of General Psychiatry* **49**, 716–722.
- Kendler, K. S. & Prescott, C. A. (1999). A population-based twin study of lifetime major depression in men and women. *Archives of General Psychiatry* **56**, 39–44.
- Kessler, R. C. (1997). The effects of stressful life events on depression. *Annual Review of Psychology* **48**, 191–214.
- Kessler, R. C., Nelson, C. B., McGonagle, K. A., Liu, J., Swartz, M. & Blazer, D. G. (1996). Comorbidity of DSM-III-R major depressive disorder in the general population: results from the US National Comorbidity Survey. *British Journal of Psychiatry* (Suppl.), 17–30.
- Maser, J. D. & Cloninger, C. R. (eds) (1990). *Comorbidity of Mood and Anxiety Disorders*. American Psychiatric Press: Washington, DC.
- Masters, T. (1993). *Practical Neural Network Recipes in C++*. Academic Press: Boston.
- Wittchen, H. U., Kessler, R. C., Pfister, H. & Lieb, M. (2000). Why do people with anxiety disorders become depressed? A prospective-longitudinal community study. *Acta Psychiatrica Scandinavica* (Suppl.), 14–23.